

When Individualized Causal Predictions Are Not Actionable: Failure Analysis, Abstention, and a Runtime-Checked Narrative Agent on Ovarian-Cancer Covariates

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Abstract

Clinical treatment-effect models can return individualized scores even when the available data do not support a stable treatment direction. We study this failure mode in a semi-synthetic benchmark built on 547 TCGA-OV covariate profiles, where treatment assignment and potential outcomes are simulated so that conditional average treatment effect (CATE) ground truth is known for evaluation without estimating the real-world effect of platinum therapy. We compare six standard CATE estimators with three matched variants of CaML-OP, an equal-weight ensemble of NonParamDML and ForestDRLearner fitted on supervised XGBoost leaf-augmented covariates. With a $K = 5$ cross-fitted leaf encoder and a four-field pre-treatment covariate set, CaML-OP ranks second in PEHE (0.0972) behind an S-Learner (0.0911); the paired difference of +0.0061 has a 95% bootstrap interval spanning zero, while MAE and absolute ATE error favor the S-Learner. CaML-OP attains the highest effect-sign accuracy of the nine methods, 0.5988, which still leaves directional recovery weak. Cross-fitting reverses the representation result: the out-of-fold encoder cuts PEHE by 47.4% relative to raw covariates (0.1848 to 0.0972, interval excluding zero) whereas the earlier leaky encoder shows only 7.4%, so a leakage-free protocol turns a gain previously reported as negligible into the largest effect in the benchmark. A repeated split-conformal study over 30 splits attains coverage 0.799, 0.902, 0.955, and 0.990 against targets 0.80, 0.90, 0.95, and 0.99, but abstention reaches 0.994 at the 0.95 target and the mean width there, 0.613, exceeds the range of the simulated effect surface. CaML-OP propagates this uncertainty into a constrained generator-verifier narrative agent, evaluated with live language models on 160 held-out states. Free text asserts a direction on 0.176 of abstention states under a lenient rubric and 0.471 under a strict one; typed tool-call output removes unsupported directional claims entirely, and two targeted retries raise the verified pass rate from 0.450 to 0.781 with 21.9% escalation on the larger backend and from 0.194 to 1.000 on the smaller. Individualized scores can therefore be computed, improved, and calibrated while remaining insufficiently supported for directional action.

1 Introduction

Heterogeneous treatment-effect estimation aims to move from population-average effects toward treatment decisions that depend on observed patient characteristics. The statistical target is usually a conditional mean contrast rather than an observed person-specific counterfactual. We therefore use the conditional average treatment effect (CATE),

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x],$$

under the potential-outcomes framework [Imbens and Rubin, 2015]. The fitted value $\hat{\tau}(x)$ is interpreted as an individualized treatment-benefit score. It is not an identified realization of $Y(1) - Y(0)$ for a particular patient.

The empirical substrate is the ovarian serous cystadenocarcinoma project of The Cancer Genome Atlas (TCGA-OV) [The Cancer Genome Atlas Research Network, 2011]. We preserve the observed covariate distribution but simulate treatment assignment and potential outcomes. This construction makes PEHE and sign recovery observable because $\tau(x)$ is known by design. It does not establish the causal effect of administering or withholding platinum therapy in the TCGA cohort.

CaML-OP tests a modeling hypothesis that is plausible in small observational datasets: prognostic structure learned from the outcome may provide useful nonlinear features for CATE estimation, and combining estimators with different orthogonalized constructions may reduce dependence on one learner. The results separate the two halves of that hypothesis. The representation half survives: an out-of-fold supervised leaf encoder reduces error substantially relative to raw covariates in every data-generating process. The decision-relevant half does not. A simple S-Learner still attains the lowest error on every accuracy metric, the PEHE gap to the improved ensemble is not resolvable at this sample size, directional recovery remains close to the reference level, and a correct split-conformal procedure reaches its nominal coverage only by producing intervals that send almost every case into abstention.

The second part of CaML-OP addresses what happens after such a failure is detected. Interval status is converted into an abstention state that is passed to a constrained narrative agent. The agent returns a typed tool call rather than unconstrained prose, and its output is released only after deterministic checks verify consistency with the model state. The paper therefore studies both a failure regime in individualized CATE estimation and a system design that prevents the communication layer from turning an unsupported direction into a confident narrative.

Contributions.

1. We document a failure regime for individualized CATE estimation on real ovarian-cancer covariates with semi-synthetic causal outcomes: the proposed ensemble does not outperform a simple baseline on any accuracy metric, and directional recovery stays near the reference level even for the best of nine methods, including after a cross-fitted encoder cuts PEHE by 47.4%—a gain the earlier leaky protocol had hidden.
2. We connect uncertainty to system behavior through abstention with a repeated split-conformal study on known semi-synthetic CATE, and report the cost of calibration in abstention rate and interval width rather than in coverage alone. The calibration factor is evaluation-only, not a transferable clinical guarantee.
3. We implement and evaluate a generator-verifier narrative agent with live language models across four conditions, using typed output, five deterministic consistency checks, targeted retry, and an explicit escalation state. Clinical validity remains a separate question.

2 Data, estimand, and evaluation

2.1 TCGA-OV covariates and semi-synthetic outcomes

After preprocessing, 547 TCGA-OV records remain. The primary analysis uses four pre-treatment fields: age at diagnosis, FIGO stage, race, and ethnicity. Race and ethnicity are one-hot encoded for every method; FIGO stage is retained as an ordinal variable because the implementation preserves the clinical stage order. The field `treatment_intent_type` used in an earlier configuration is excluded, because it originates from treatment metadata

rather than a guaranteed pre-treatment baseline measurement. The main conclusions are methodological and concern the behavior of estimators on this empirical feature distribution.

Four data-generating processes preserve the empirical X distribution while varying the CATE surface from constant to linear, nonlinear, and interaction forms. Treatment assignment is generated from a logistic function of standardized age and stage, and potential outcomes are simulated so that $\tau(x)$ is known. Each scenario is evaluated over 10 seeds with a stratified 75/25 train-test split, yielding approximately 410 training and 137 test observations per run. The assignment model is a benchmark construction, not a reconstruction of actual TCGA treatment allocation.

At the baseline assignment setting, propensity scores span 0.164–0.872, so trimming the training fold at 0.05–0.95 removes no observations. The main benchmark therefore studies confounding under comfortable overlap rather than severe positivity violations.

2.2 Baselines and metrics

The comparison includes S-, T-, and X-Learners [Künzel et al., 2019], a causal-forest baseline in the tradition of Wager and Athey [2018], NonParamDML using residualized nuisance estimation within a double-machine-learning/R-learner family [Chernozhukov et al., 2018, Nie and Wager, 2021], and ForestDRLearner, a forest-based doubly robust CATE estimator related to the pseudo-outcome construction analyzed by Kennedy [2023]. CaML-OP is run as three variants on identical splits and seeds: (A) raw covariates, (B) the earlier leaky leaf representation, and (C) the cross-fitted leaf representation used as the primary result. Semi-synthetic CATE benchmarks are useful because the effect surface is observed by construction, but method rankings can depend on the chosen DGP and inductive bias [Curth et al., 2021]; this dependence is part of the failure analysis rather than a nuisance to be hidden.

We report PEHE, mean absolute effect error, absolute ATE error, effect-sign accuracy where the simulated effect surface contains both signs, and empirical interval coverage, and summarize method differences with paired 95% bootstrap intervals over the $4 \times 10 = 40$ (DGP, seed) pairs. Off-policy values are not used to support the main conclusions because the evaluated sign policy does not match the abstention-aware behavior of the system.

3 CaML-OP

3.1 Representation and CATE ensemble

A gradient-boosted classifier is fit to the binary outcome, and each tree’s leaf assignment is one-hot encoded. These leaf indicators are concatenated with the original covariates to form a supervised basis expansion using XGBoost [Chen and Guestrin, 2016]. NonParamDML and ForestDRLearner are then fit on the augmented design, and their point CATE predictions are averaged with equal weight.

Equal weighting is an empirical ensemble choice; averaging the two estimators does not create an additional robustness theorem. The leaf encoder is cross-fitted so that it is out-of-fold with respect to the estimation stage. Training leaf features come from $K = 5$ fold-wise fitting: for each fold the encoder is fit on the remaining $K - 1$ folds and transforms the held-out fold, and the held-out blocks are concatenated into the training representation. For calibration and evaluation observations the encoder is fit on model-training data only and applied as a transform, so no calibration or evaluation outcome enters the representation. Variant B retains the earlier protocol, in which the encoder sees training outcomes before the causal estimators perform their internal cross-fitting, and is reported only as a leakage comparison.

3.2 Split-conformal interval and abstention

The equal-weight ensemble has no native confidence interval. CaML-OP forms an uncalibrated interval by taking the ForestDR Learner interval width, re-centering it on the ensemble point estimate, and multiplying the half-width by 1.10. This heuristic half-width h is then calibrated by a split-conformal procedure rather than used as a nominal 95% interval.

Each repetition partitions the observations into model-training, calibration, and evaluation sets before any fitting, so the pipeline, including the cross-fitted training representation, is fit on model-training data alone. On the calibration set we compute scores $s_i = |\tau_i - \hat{\tau}_i| / \max(h_i, \epsilon)$ and take the k -th order statistic with $k = \lceil (n+1)(1-\alpha) \rceil$ clipped to $[1, n]$. The threshold is frozen, and coverage, width, and abstention are measured only on the untouched evaluation set. If the calibrated interval around $\hat{\tau}(x)$ spans zero, the system enters an abstention state and the agent may not assert a treatment direction.

This is a semi-synthetic, oracle split-conformal study: the scores use known $\tau(x)$, which is unobservable for real patients, so the resulting threshold is not transferable to a clinical deployment. Conformal approaches to counterfactual and treatment-effect uncertainty require explicit causal assumptions and treatment of the missing-counterfactual problem [Lei and Candès, 2020]. We make no subgroup-conditional coverage claim, because no method carrying that guarantee is implemented here.

3.3 Effect-function attribution and hidden-confounding diagnostic

The fitted CATE score is wrapped as a prediction function and attributed over the four raw fields with model-agnostic permutation SHAP [Lundberg and Lee, 2017]. The attributions describe how the fitted $\hat{\tau}(x)$ changes relative to its background expectation; they are not causal effects of the covariates themselves. Treatment-targeted or treatment-effect-specific explanation methods already exist, including PWSHAP [Ter-Minassian et al., 2023], DISCRET [Wu et al., 2024], and PermuCATE [Paillard et al., 2025]. We therefore use the narrower term *effect-function attribution* rather than introducing a new causal explanation method.

The implementation also applies the classical E-value transformation to model-predicted arm risks. The E-value was introduced for sensitivity analysis of treatment–outcome associations on the risk-ratio scale [VanderWeele and Ding, 2017]; a local transformation of two predicted arm risks is not the same estimand. We use it only as an E-value-inspired robustness diagnostic. Formal analyses of heterogeneous effects under hidden confounding include ignorance intervals [Jesson et al., 2021] and B-Learner bounds [Oprescu et al., 2023].

Figure 1 summarizes the separation between runtime quantities and semi-synthetic ground truth used only for evaluation.

3.4 Runtime-checked narrative agent

The narrative agent receives a structured patient state rather than raw clinical records. The state contains the estimated CATE score, calibrated interval, abstention flag, estimated arm-level outcomes, effect-function attributions, model disagreement, and the exploratory robustness diagnostic. The generator must return a typed tool call containing a narrative, a direction state, cited features, tagged numerical claims, and a declared dominant uncertainty source.

Five deterministic checks are applied before release: (1) direction/abstention consistency, (2) membership of cited features in the supplied attribution set, (3) consistency of tagged numerical claims with the corresponding model quantities, (4) consistency between declared direction and the ordering of estimated arm outcomes, and (5) agreement between the declared dominant uncertainty source and a deterministic ranking of the supplied diagnostic signals. A failed check produces targeted feedback and another generation attempt; exhaustion of two retries yields an **ESCALATED** state for human review. The checks verify consistency with model outputs. They do not certify medical correctness.

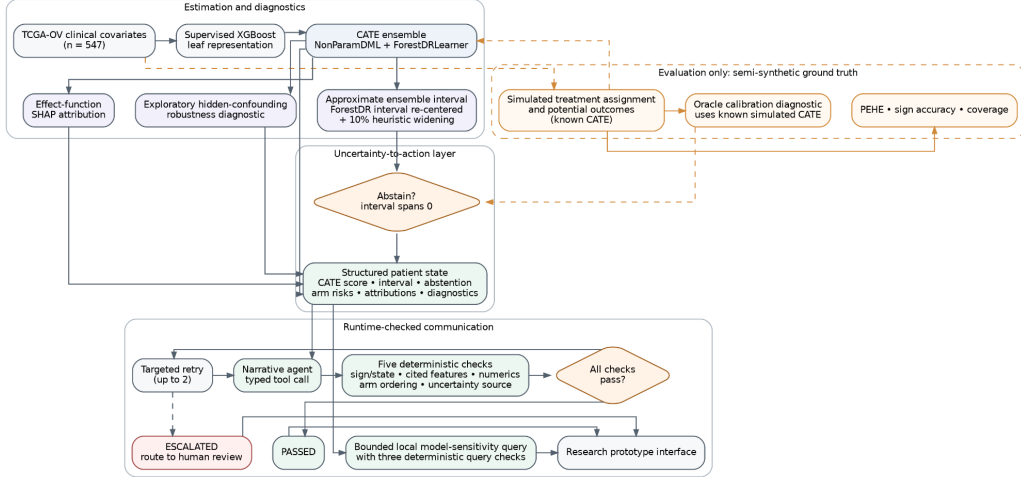


Figure 1: CaML-OP architecture. Dashed orange components use semi-synthetic ground truth and are available only during evaluation, including the split-conformal calibration scores. The runtime path propagates the CATE score, calibrated interval, abstention state, arm-level predictions, attributions, and diagnostics to a constrained narrative agent.

Table 1: Mean estimation error over four DGPs and ten runs per DGP, four-field pre-treatment covariate set. Lower is better for PEHE, MAE, and ATE error. Sign accuracy is averaged over the DGPs whose effect surface contains both signs; DGP 1 has a constant effect and is excluded from that column.

Method	PEHE	MAE	ATE error	Sign acc.
S-Learner	0.0911	0.0712	0.0357	0.5760
CaML-OP [C, cross-fitted]	0.0972	0.0792	0.0555	0.5988
Causal Forest	0.1326	0.1101	0.0546	0.5685
CaML-OP [B, leaky]	0.1711	0.1318	0.0564	0.5517
X-Learner	0.1734	0.1397	0.0519	0.5879
CaML-OP [A, raw]	0.1848	0.1470	0.0585	0.5779
T-Learner	0.2202	0.1751	0.0490	0.5636
R-Learner	0.2612	0.2039	0.0628	0.5670
DR-Learner	0.2752	0.2116	0.0735	0.5617

The interface also provides a bounded local model-sensitivity query. A selected input field is perturbed and the fitted models are re-evaluated at the modified input. Without an explicit structural causal model that gives the perturbation intervention semantics, this operation should not be interpreted as a patient counterfactual [Pearl, 2009].

4 Results

4.1 Estimation accuracy

Table 1 summarizes the primary benchmark. The cross-fitted CaML-OP variant ranks second of nine on PEHE and MAE, ahead of the causal forest and of both other CaML-OP variants, and behind the S-Learner. The three orthogonalization-based baselines occupy the bottom of the ranking.

Paired 95% bootstrap intervals over the 40 (DGP, seed) pairs sharpen this ordering. Variant C lowers PEHE by 0.0875 (CI -0.0962 to -0.0787) against raw covariates and by

0.0739 (CI -0.0843 to -0.0635) against the leaky variant, with MAE differences of the same sign and significance; in both contrasts the ATE-error interval contains zero. Against the S-Learner, the PEHE difference of $+0.0061$ (CI -0.0022 to $+0.0145$) also contains zero, so the two are not distinguishable on that metric, whereas MAE $+0.0081$ (CI $+0.0006$ to $+0.0157$) and ATE error $+0.0198$ (CI $+0.0091$ to $+0.0302$) favor the S-Learner. The representation therefore improves the shape of the estimated effect surface without improving the average effect it implies.

4.2 Effect direction and signal strength

Effect-sign accuracy is the decision-relevant quantity, and it separates the methods far less than PEHE does: all nine lie between 0.5517 and 0.5988 (Figure 2, left). The cross-fitted variant is highest at 0.5988, ahead of the X-Learner at 0.5879 and the S-Learner at 0.5760; the leaky variant is lowest at 0.5517. A 47.4% reduction in PEHE thus buys 2.1 points of sign accuracy over raw covariates. The dashed 0.5 line is a reference level, not an estimated majority-class baseline, since sign balance is not computed per DGP. The earlier effect-scale sweep was run under the superseded five-field configuration and is not carried into the primary results.

4.3 Assignment strength and representation ablation

The assignment-strength sweep, which sharpens the assignment mechanism through a multiplier c , was run under the superseded five-field configuration with the leaky encoder and has not been repeated, so it is not reported as a primary result. Its qualitative observation still applies to any rerun: the multiplier changes confounding strength and overlap simultaneously and so cannot identify which mechanism drives a degradation.

The representation ablation is consistent within every DGP. Measured in PEHE, the cross-fitted representation improves on raw covariates from 0.1795 to 0.0862 in the constant DGP (52.0%), 0.1892 to 0.1014 in the linear DGP (46.4%), 0.1828 to 0.0921 in the nonlinear DGP (49.6%), and 0.1876 to 0.1091 in the interaction DGP (41.8%), for an overall reduction of 47.4%. The leaky representation yields only 7.4% overall and is beaten in every DGP.

This reverses the conclusion reported under the earlier protocol, where the leaf representation appeared to contribute about 5% and was described as within hyperparameter noise. The gain was not small, it was obscured: the encoder being compared against raw covariates had been fit on the outcomes it was then evaluated on. The negative result here concerns the evaluation protocol rather than the representation.

4.4 Coverage and abstention

Thirty independent repetitions of the split-conformal procedure on DGP 4 use calibration sets of 106 and evaluation sets of 107 observations. Achieved coverage matches the nominal target at every level: 0.799 against 0.80, 0.902 against 0.90, 0.955 against 0.95, and 0.990 against 0.99, and the 95% interval across repetitions contains the target in all four cases (Figure 2, right).

Coverage is the least informative of the three quantities. Abstention, the share of evaluation cases whose calibrated interval spans zero, rises from 0.850 at the 0.80 target to 0.994 at 0.95 and 1.000 at 0.99, and mean width rises in step from 0.323 through 0.450 and 0.613 to 0.984; at the 0.95 target the mean width already exceeds the range of the simulated effect surface. Correcting the calibration procedure therefore strengthened rather than softened the earlier qualitative claim: a procedure that delivers its nominal coverage here is one that declines to state a direction for essentially every patient. Intervals across repetitions are wide at the lower targets, for instance 0.683 to 0.918 for coverage at the 0.80 target, which is what a calibration set of 106 supports.

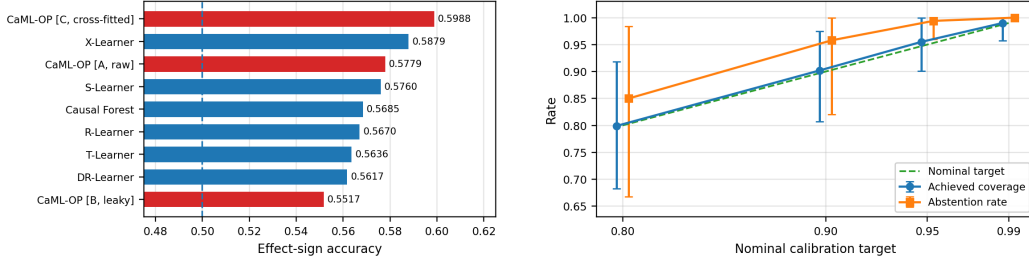


Figure 2: Primary semi-synthetic diagnostics. Left: effect-sign accuracy by method in the primary four-field benchmark; CaML-OP variants are shaded, and the dashed 0.5 line is a reference level, not an estimated majority-class baseline. Right: repeated split-conformal calibration on DGP 4 over 30 independent splits; markers are means and bars are 95% intervals across repetitions, with known simulated CATE used only for evaluation.

Table 2: Live-LLM evaluation over 160 held-out states. Condition A is scored by parsing free text under the lenient rubric of Section 4.5; the strict rubric gives 0.471 for the Sonnet backend, and the Haiku free-text condition was not rescored with the corrected parser. Released-unsupported rate equals the unsupported-direction rate in condition A and is 0.000 in B, C, and D.

Backend	Cond.	Unsup. dir.	Numeric err.	Verified pass	Escalation	Calls/case
Sonnet 5	A	0.176	0.000	—	0.000	1.00
Sonnet 5	B	0.000	0.000	0.450	0.000	1.00
Sonnet 5	C	0.000	0.000	0.450	0.550	1.00
Sonnet 5	D	0.000	0.000	0.781	0.219	2.14
Haiku 4.5	A	n/a	0.000	—	0.000	1.00
Haiku 4.5	B	0.000	0.469	0.194	0.000	1.00
Haiku 4.5	C	0.000	0.469	0.194	0.806	1.00
Haiku 4.5	D	0.000	0.000	1.000	0.000	2.59

4.5 Verifier and retry behavior under live models

The agent is evaluated on 160 held-out structured patient states, stratified across score sign, interval width, model disagreement, and robustness warning; 93 require abstention and 67 do not. Two backends of different size from one model family are used: `claude-sonnet-5` at the model default temperature and `claude-haiku-4-5-20251001` at temperature 0.0. Four conditions run on identical states: (A) free-text baseline, (B) typed tool-call output without the gate, (C) typed output with the five checks, and (D) typed output with checks and up to two targeted retries (Table 2).

When generation is unconstrained, both backends assert a treatment direction on states where the model is in abstention. Moving to a typed tool call removes that failure mode outright, before any gate is applied: the unsupported-direction rate is 0.000 in conditions B, C, and D for both backends. The remaining failures differ by backend. Haiku misstates tagged numbers on 46.9% of states in condition B and clears all five checks on 19.4%; two retries eliminate the numeric errors and bring the verified pass rate to 1.000 with no escalations. Sonnet produces no numeric errors in any condition, yet clears all five checks on only 45.0% of states in B, rising to 78.1% in D, with 21.9% of cases escalating after two retries. Escalation in C equals one minus the condition-B pass rate by construction. Condition D costs 2.14 calls per case for Sonnet and 2.59 for Haiku, at 12.4 s and 9.9 s mean latency and a total spend of \$5.92 and \$1.76.

The gate is separately stress-tested on the existing 15-case regression suite and on 128 generated cases, 14 for each of the eight implemented failure classes plus 16 faithful controls. Sensitivity and specificity are 1.000 on both: every corrupted case fails at least one check and

every faithful case passes all five, which establishes coverage of the encoded failure classes, not completeness of the verifier. As a control-loop mechanism test, injecting a 30% per-call corruption probability over 1,000 trials gives pass rates of 0.736, 0.930, and 0.978 at zero, one, and two retries, at 1.345 calls on average; that synthetic process is not a substitute for the live-model rates above. VeriFact is a related example of verifying LLM-generated clinical text against health records, with a different task and mechanism [Chung et al., 2025].

Condition A carries a measurement caveat that is itself a finding. Free text has no typed direction field, so it must be scored by parsing narratives; our initial parser matched the bare substring “benefit” and, on a 30-state audit, marked 29 of 30 narratives as positive assertions, four of which asserted a negative effect. A rewritten parser handling negation and hedging gives an abstention-violation rate of 0.176 under a lenient rubric, which exempts narratives that report a point estimate, state that the interval spans zero, and make no recommendation, and 0.471 under a strict rubric counting any signed statement; the wrong-direction rate on non-abstention states is 0.000 under both. That a rubric choice moves the headline number by nearly a factor of three on identical text is why the typed-output comparison is the load-bearing one.

5 Failure analysis and boundary conditions

The negative results arise from several interacting conditions. The DGPs are relatively smooth, and semi-synthetic benchmark design can favor particular inductive biases [Curth et al., 2021]; this is a plausible explanation for the S-Learner’s continued strength, not a mechanism identified here. Sign accuracy near the reference level across all nine methods, despite a 47.4% spread in PEHE, indicates limited directional signal rather than a deficiency specific to any estimator. Correct conformal calibration converts that missing signal into interval width, and width into near-total abstention. The four-field clinical representation is also too sparse to support strong real-world ignorability claims, irrespective of predictive accuracy.

These limitations also define the role of the narrative agent. Deterministic checks can prevent generated text from contradicting an abstention state, citing unavailable features, reporting inconsistent numbers, or reversing the estimated arm ordering. They cannot transform a weakly supported causal quantity into a clinically valid one. CaML-OP therefore separates estimation from communication: the causal layer determines what the model supports, and the verification layer constrains what the agent is allowed to say about it.

Table 3 summarizes the observed failures and the boundary conditions they expose.

6 Related work

The causal estimand follows the potential-outcomes framework [Imbens and Rubin, 2015]. The estimation backbone is related to established meta-learners [Künzel et al., 2019], causal forests [Wager and Athey, 2018], double/debiased machine learning [Chernozhukov et al., 2018], R-learner estimation [Nie and Wager, 2021], and doubly robust CATE estimation [Kennedy, 2023]. The cross-fitting protocol applied to the representation stage follows the same logic that motivates cross-fitting of nuisance functions [Chernozhukov et al., 2018]. The benchmark interpretation follows the caution of Curth et al. [2021], who show that semi-synthetic CATE evaluations can systematically favor particular methods. XGBoost supplies the supervised tree representation used here [Chen and Guestrin, 2016].

For interpretation, standard SHAP attributes a fitted prediction function [Lundberg and Lee, 2017]. PWSHAP develops path-wise Shapley effects for a targeted variable using a user-specified DAG [Ter-Minassian et al., 2023]; DISCRET constructs faithful rule-based explanations for treatment-effect estimation [Wu et al., 2024]; and PermuCATE provides statistically grounded global variable importance for CATE heterogeneity [Paillard et al., 2025].

Table 3: Observed failure modes and their empirical boundary conditions.

Finding	Evidence	Boundary condition / implication
Simple baseline not outperformed	PEHE 0.0911 (S-Learner) vs. 0.0972 (CaML-OP [C]); paired PEHE CI spans zero, MAE and ATE CIs favor the S-Learner	Added architecture buys a tie on PEHE and a loss elsewhere; DGP inductive bias matters.
Directional recovery is limited	Sign accuracy 0.5517–0.5988 across all nine methods	Reliable direction needs stronger signal or a different decision target.
Calibration achieved only through abstention	Coverage 0.955 at the 0.95 target, abstention 0.994, mean width 0.613	Nominal coverage is attainable and uninformative here; the oracle threshold is not transferable.
Leakage hid a real representation gain	Cross-fitted encoder cuts PEHE 47.4% vs. raw; leaky encoder shows 7.4%	A leaky ablation understated the representation; protocol errors are not always self-serving.
Free text cannot be scored stably	Abstention violations 0.176 (lenient) vs. 0.471 (strict) on identical text	Typed fields are what make the safety claim measurable.
Verification bounds behavior, not validity	Unsupported direction 0.000 in B–D; verified pass 0.450→0.781 (Sonnet) and 0.194→1.000 (Haiku)	Checks enforce model-state consistency; escalation is the honest output when the state is unsupported.

These methods motivate our deliberately narrow interpretation of SHAP as attribution of the learned effect function.

For uncertainty and hidden confounding, Lei and Candès [2020] develop conformal inference for counterfactuals and treatment effects under explicit causal assumptions. The E-value is a risk-ratio-scale sensitivity measure for unmeasured confounding in treatment–outcome associations [VanderWeele and Ding, 2017], whereas Jesson et al. [2021] and Oprescu et al. [2023] directly study intervals or bounds for heterogeneous effects under hidden confounding. For the communication layer, VeriFact verifies factual support of generated clinical text against EHR evidence [Chung et al., 2025]. Our verifier is narrower: it deterministically checks consistency with quantities produced by the causal pipeline. The distinction between model-sensitivity perturbations and causal interventions follows standard structural-causal semantics [Pearl, 2009].

7 Limitations

The study is a methodological semi-synthetic benchmark, not a clinical validation. Treatment assignment and potential outcomes are simulated, so no real-world platinum-treatment effect is identified. Only four pre-treatment fields are available, which is too sparse to support a credible real-world adjustment set. Baseline overlap is comfortable by construction, so the benchmark does not probe positivity violations. The conformal calibration is an oracle procedure that uses known semi-synthetic $\tau(x)$; the resulting multiplier cannot be computed for real patients and no subgroup-conditional coverage is claimed. Calibration sets of 106 observations leave visible variation between repetitions. The two language-model backends differ in size but come from a single model family, so cross-provider generalization of the agent results is untested. The free-text baseline is scored by a text parser under two rubrics rather than by clinician judgment, and the free-text condition was not rescored

for the smaller backend. No clinician has assessed the narratives, abstention behavior, or interface. These limitations constrain the claims to estimator behavior, uncertainty-aware abstention, and model-state consistency within the benchmark studied here.

8 Conclusion

The central empirical result is that a substantial modeling improvement can leave the decision unchanged. Cross-fitting the supervised representation cuts PEHE by 47.4% and lifts CaML-OP to second of nine methods, statistically inseparable from the S-Learner on PEHE, yet effect-sign accuracy moves only from 0.5779 to 0.5988 and a correctly calibrated interval abstains on 0.994 of evaluation cases at the 0.95 target. With a small sample, sparse clinical information, and limited heterogeneous signal, an individualized CATE score can be improved, calibrated, and still provide no reliable basis for directional action.

CaML-OP makes this distinction operational. An abstention state blocks directional narration when the interval spans zero, and the generator-verifier agent must remain consistent with the model state before output is released. Live-model evaluation shows what that buys: typed output eliminates unsupported directional claims that free text produces on 17.6% to 47.1% of abstention states depending on the rubric, and targeted retry converts most remaining check failures into either a verified narrative or an explicit escalation. This does not repair identification, calibration, or clinical validity; it prevents the communication layer from hiding those limitations. For causal decision-support systems, recognizing when the evidence does not support a directional claim is a distinct capability from estimating the score itself.

Reproducibility, ethics, and LLM-use disclosure

The reported experiments use TCGA-OV covariates with semi-synthetic treatment and outcome mechanisms. The study does not provide treatment advice. Code and supplementary material must remain anonymous during double-blind review. All numerical results reported here are taken from the supplied experimental outputs. Large language model tools were used for drafting support, language editing, and manuscript organization; they were not used to generate experimental results or determine the statistical conclusions reported here. The authors are responsible for the final text, numerical results, figures, and references. The language models used within the CaML-OP narrative agent are part of the method, are named with version identifiers in Section 4.5, and are described in Section 3.4.

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